MODULE 34 · EXAM STRATEGY - ALL DOMAINS

Confusion Pairs

Sixty services that look like each other, and the one question that separates each pair.

DOMAIN Cross-cutting EXAM WEIGHT All MODULE 34 of 35

admin@sthithapragna.com

Every confusing pair reduces to **one question**. Find the question and you have the answer, without recalling the full comparison.

Monitoring and logging

PAIR	THE ONE QUESTION
Basic or detailed monitoring	Do you need one-minute EC2 metrics?
Standard or high resolution	Is the metric published more often than once a minute?
Average or percentile	Are you hiding a slow minority behind a healthy mean?
EMF or PutMetricData	Is the metric derived from something you are already logging?
Metric filter or Logs Insights	Forward-looking alarm, or backward-looking question?
Subscription filter or export to S3	Streaming continuously, or a bulk copy?

ON THE EXAM

The metric filter versus Logs Insights split is the most useful here. A filter creates a metric going forward; Logs Insights queries what is already stored.

Alarms, events and remediation

PAIR	THE ONE QUESTION
Missing as breaching or not breaching	Is silence a symptom, or is the resource simply idle?
Static threshold or anomaly detection	Do you know the right number, or only the normal shape?
Composite alarm or mute rule	Suppress by logic, or suppress by calendar?
Scheduled rule or EventBridge Scheduler	Do you need time zones, DST or a one-time schedule?
EventBridge or SNS	Route on event content, or fan out to subscribers?
Config remediation or EventBridge remediation	Drifted existing state, or a fresh event?

ON THE EXAM

Config finds things that were already wrong before the rule existed. EventBridge only ever sees what happens next. That is the whole distinction.

Systems Manager and operations

PAIR	THE ONE QUESTION
Run Command or State Manager	Once now, or continuously enforced?
Session Manager or EC2 Instance Connect	Do you need a fully logged, port-free session?
Parameter Store or Secrets Manager	Does it need automatic rotation?
Automation or Run Command	Acting on the AWS resource, or inside the operating system?
Patch policies or maintenance windows	Are you building patching new, or maintaining what exists?
Runbook or DevOps Agent	Is the cause known, or does it need investigating?

ON THE EXAM

Run Command versus State Manager is the same question as EventBridge versus Config: one acts once, the other keeps a state true.

Compute and storage performance

PAIR	THE ONE QUESTION
Standard or unlimited credit mode	Predictable bill, or predictable performance?
gp2 or gp3	Almost always gp3. Is there any reason not to?
gp3 or io2 Block Express	Above 80,000 IOPS, sub-millisecond, or five-nines durability?
Volume limit or instance limit	Did throughput halve about 30 minutes in?
Transfer Acceleration or DataSync	One distant client, or a file server's worth of data?
DAX or ElastiCache	Is the database DynamoDB, and must code changes be minimal?

ON THE EXAM

The volume-versus-instance row is not a service pair at all, but it is where storage performance questions are actually decided.

Databases and scaling

PAIR	THE ONE QUESTION
Multi-AZ or read replica	Availability, or read scale?
Multi-AZ instance or Multi-AZ cluster	Do you need the standbys to serve reads?
Provisioned Aurora or Aurora serverless	Is the load steady, or spiky and unpredictable?
DynamoDB on-demand or provisioned	Can you predict the load?
Node-based or serverless ElastiCache	Do you want control, or do you want it to just work?
Scale the database or add a cache	Do a few items dominate reads, and is staleness tolerable?

ON THE EXAM

Multi-AZ versus read replica is the single most confused pair in the RDS syllabus: availability versus scale, synchronous versus asynchronous, automatic versus manual.

Load balancing and traffic

PAIR	THE ONE QUESTION
ALB or NLB	Do you need HTTP routing, or static IPs and any protocol?
Target tracking or step scaling	Do different severities need different responses?
Evaluate target health or a health check	Is the record an alias to a load balancer?
502 or 503	Did a target answer badly, or was there no target?
503 or 504	Was there no target, or did it not answer in time?
Cross-zone on or off	ALB is always on and free. NLB is off and charged

ON THE EXAM

The error-code rows are the highest-value entries in this whole module. 502 answered badly, 503 nobody answered, 504 nobody answered in time.

Reliability and recovery

PAIR	THE ONE QUESTION
Pilot light or warm standby	Is anything running and serving in the second Region?
Automated backups or manual snapshots	Must it survive the instance being deleted?
Vault Lock governance or compliance	Does the requirement say "including the root user"?
Zonal shift or routing controls	One impaired Availability Zone, or a whole Region?
Cross-Region copy or cross-account copy	Defending against a Regional event, or a compromised account?
Warm pool or a better AMI	Is the boot time spent installing, or loading current data?

ON THE EXAM

"Including the root user" or "including administrators" is the discriminator for compliance mode, and it appears in Object Lock, Vault Lock and CloudTrail questions alike.

Deployment and provisioning

PAIR	THE ONE QUESTION
DeletionPolicy or UpdateReplacePolicy	Is the resource leaving the stack, or being replaced?
Ref or GetAtt	Do you want the default identifier, or a named attribute?
Nested stacks or cross-stack exports	Who owns the lifecycle: one team, or several?
Self-managed or service-managed StackSets	Must accounts created later be covered automatically?
CDK or plain CloudFormation	Does the infrastructure have real logic, or is the template the artefact?
Terraform or CloudFormation	Does it span providers, or is there an existing Terraform estate?

ON THE EXAM

DeletionPolicy and UpdateReplacePolicy is the most expensive pair in this table. Both default to Delete, and setting only one leaves a database unprotected.

Containers and deployment strategy

PAIR	THE ONE QUESTION
Task role or task execution role	Did it fail before your code ran?
ECS on EC2 or Fargate	Do you need specific instance types, or no hosts at all?
ECR basic or enhanced scanning	Must a clean result stay true as new CVEs appear?
IRSA or EKS Pod Identity	Both give a pod its own role. Is a per-cluster OIDC provider acceptable?
Immutable or blue/green	Does the old environment survive the switch?
Rolling or canary	Do you need a bake period before everyone is exposed?

ON THE EXAM

Task role versus execution role is decided by timing, not by memorising a list. Anything happening before your code runs belongs to the execution role.

Serverless and orchestration

PAIR	THE ONE QUESTION
Reserved or provisioned concurrency	Is the complaint throttling, or slow first requests?
Pool limit or scaling rate	Were you throttled with capacity still free?
Destinations or a DLQ	Do you need the request and response context?
Step Functions Standard or Express	Long and auditable, or short and high volume?
Step Functions or EventBridge Pipes	Orchestrating many steps, or connecting one source to one target?
S3 notifications or EventBridge	Do several consumers need overlapping filters?

ON THE EXAM

The second row is not a service pair. It is the Lambda behaviour almost nobody teaches: two independent limits, and only one of them is on the dashboard.

Identity and governance

PAIR	THE ONE QUESTION
Permissions boundary or SCP	Capping one identity, or every principal in an account?
SCP or RCP	Is the risk your people acting, or someone reaching your data?
Declarative policy or SCP	Enforcing a setting, or denying an action?
Identity Center or IAM users	Must leavers be deprovisioned everywhere at once?
RAM or a cross-account role	Sharing a thing, or sharing an identity?
Key policy or grant	Durable and declared, or temporary and programmatic?

ON THE EXAM

SCP versus RCP is the newest pair here and the most likely to appear. Ask who is acting: your engineers, or somebody outside reaching in.

Detection and encryption

PAIR	THE ONE QUESTION
Inspector or GuardDuty	A vulnerability that could be exploited, or activity happening now?
Macie or GuardDuty S3 Protection	What is inside the objects, or how they are accessed?
Config or CloudTrail	The resulting state, or the API call that caused it?
Hooks or Config rules	Block it before provisioning, or find it afterwards?
Parameter Store or Secrets Manager	Does it need automatic rotation?
SSE-S3 or SSE-KMS	Must key usage be auditable and the policy yours?

ON THE EXAM

Inspector finds the unlocked door; GuardDuty sees somebody walk through one. Macie reads the contents; GuardDuty watches the behaviour.

Networking

PAIR	THE ONE QUESTION
Security group or network ACL	Stateful or stateless. Is return traffic the problem?
Gateway or interface endpoint	Must on-premises reach it?
Peering or Transit Gateway	Two VPCs, or many plus hybrid?
PrivateLink or peering	Do the CIDR ranges overlap?
Direct Connect or VPN	Consistent and high volume, or fast to stand up?
Alias or CNAME	Is it the zone apex?

ON THE EXAM

PrivateLink versus peering is decided by one clause: overlapping CIDRs. PrivateLink does not route, so overlaps are irrelevant to it.

DNS and the edge

PAIR	THE ONE QUESTION
Geolocation or geoproximity	Where the user is, or how far from your sites?
Inbound or outbound Resolver endpoint	Which direction does the DNS query travel?
Latency or geolocation routing	Measured network performance, or the user's country?
CloudFront or Global Accelerator	Is the content cacheable HTTP?
CloudFront Functions or Lambda@Edge	Does it need to call something?
Shield Standard or Advanced	Do you need AWS specialists during an attack, and cost protection?

ON THE EXAM

CloudFront versus Global Accelerator is the most reliably examined pair in Domain 5. Global Accelerator is not a cache, and it handles any protocol.

Three-way confusions

TRIO	HOW TO SPLIT THEM
CloudTrail, Config, CloudWatch	Who did it. What it looks like now. How it is performing
SQS, SNS, EventBridge	Decouple in time. Decouple in number. Decouple in content
Reachability Analyzer, Network Access Analyzer, flow logs	This path? All paths? What actually flowed?
Preventive, detective, corrective	A Hook blocks it. Config finds it. Automation fixes it
Inspector, GuardDuty, Macie	Vulnerability. Behaviour. Data contents
ASG, target group, Route 53 health checks	Replace it. Stop sending to it. Stop resolving to it

ON THE EXAM

The last row catches more people than any pair in this module. Three services use the phrase health check for three different verbs.

One word, different meanings

WORD	WHAT IT MEANS WHERE
Health check	Replace an instance, stop routing to a target, or change a DNS answer
Endpoint	A VPC endpoint, a Resolver endpoint, a service URL, or a PrivateLink service
Policy	IAM policy, key policy, stack policy, scaling policy, backup policy, RCP
Grant	A KMS grant, or an IAM Allow. Only one is temporary and programmatic
Warm	Warm pool, warm standby, warm throughput. Three unrelated ideas
Rotation	KMS key material, a Secrets Manager credential, or log rotation

ON THE EXAM

When a question uses one of these words, read the surrounding sentence for which meaning applies. The word alone tells you nothing.

Pairs where the answer is both

- **Multi-AZ and read replicas.** Availability and read scale are different requirements, and most production databases need both.
- **Predictive and target tracking scaling.** Predictive only scales out, so target tracking handles scale-in and surprises.
- **Direct Connect and VPN.** The standard design is Direct Connect primary with a VPN as automatic backup.
- **Inspector and GuardDuty.** One finds weaknesses, the other finds activity, and GuardDuty's newer findings use Inspector's context.

> ON THE EXAM
>
> A question forcing a choice between these usually adds a constraint that rules one out. Without such a constraint, treat "both" as the honest answer.

Pairs where the newer option is now the default

- **OAC over OAI. Launch templates over launch configurations. ADOT over the X-Ray SDK.**
- **gp3 over gp2. Instance refresh over AutoScalingRollingUpdate. ECS-native blue/green over CodeDeploy.**
- **Identity Center over IAM users. Destinations over Lambda DLQs. Valkey over Redis OSS** for new ElastiCache deployments.
- In each case the older option still works and still appears in answer options. It is the dated answer, not an incorrect one.

> ON THE EXAM
>
> These are the pairs where an option can be technically workable and still wrong. The question is usually which is recommended, not which is possible.

> A question describes an application where server-side metrics are healthy, the load balancer reports normal latency, and a small group of users in one country report slowness. It asks how to determine what those users experienced.
>
> The options include CloudWatch Synthetics, CloudWatch RUM, X-Ray, and detailed monitoring.

Which pair is being tested, and what decides it?

A Synthetics or RUM: does the answer need real users, or robots on a schedule?

B X-Ray or Application Signals: one request, or one service over time?

C Basic or detailed monitoring: is one-minute granularity needed?

D Static threshold or anomaly detection

◆ PAUSE AND DECIDE

Which pair is being tested, and what decides it?

A Synthetics or RUM

The question asks what real users experienced. Synthetics runs robots from AWS; only RUM reports actual browsers, by geography and device.

B X-Ray or Application Signals

Both stop at the server, and every server-side measurement is stated as healthy. Neither can see the client side.

C Basic or detailed monitoring

Changes how often EC2 metrics are collected. More frequent healthy metrics are still healthy metrics.

D Threshold or anomaly detection

About how an alarm decides to fire. Nothing here concerns alarm configuration.

THE SIGNAL

Healthy server, unhappy users is the gap only RUM measures. Recognising which pair a question is testing is often faster than evaluating four services.

> A question requires that seven years of records must not be deletable by anyone in the company, for the full retention period. That explicitly includes administrators and the root user.
>
> Options include Object Lock governance mode, Object Lock compliance mode, MFA delete, and a restrictive bucket policy.

What decides this?

A The phrase "including the root user" eliminates every control a privileged user can reverse

B The seven-year duration, which only compliance mode supports

C Whether versioning is already enabled on the bucket

D Whether the records are encrypted

◆ PAUSE AND DECIDE

What decides this?

A "Including the root user"

Governance mode, MFA delete and a bucket policy can all be reversed by someone with sufficient privilege. Compliance mode cannot, by anyone.

B The seven-year duration

Governance mode supports any retention period too. Duration is not what separates the two modes.

C Versioning enabled

A prerequisite for Object Lock in either mode, so it cannot discriminate between them.

D Encryption

Unrelated. Encryption controls who can read the data, not who can delete it.

THE SIGNAL

That phrase is the discriminator in Object Lock, Vault Lock and CloudTrail questions alike. It always eliminates the reversible option.

A question describes an Auto Scaling group behind a load balancer where instances launch, are marked unhealthy, are terminated and replaced, repeatedly. The application takes several minutes to start.

Options mention health check grace period, instance warm-up, cooldown, and deregistration delay.

Which timer is being tested?

A The health check grace period, because it governs when health checks may judge a new instance

B The instance warm-up, because it governs when a new instance counts

C The cooldown, because it governs the gap between scaling activities

D The deregistration delay, because it governs draining

◆ PAUSE AND DECIDE

Which timer is being tested?

A Health check grace period

It is the only timer that decides when health checks may terminate a new instance. Set below start-up time, it creates a permanent replacement loop.

B Instance warm-up

The closest distractor. Warm-up excludes a booting instance from scaling metrics; it never causes or prevents a termination.

C Cooldown

Pauses further scaling activity after one completes. It has no bearing on health evaluation.

D Deregistration delay

Governs how long a removed target keeps draining. It applies on the way out, not on the way in.

THE SIGNAL

Four timers, and only one can terminate an instance. Warm-up hides metrics, cooldown blocks scaling, deregistration delay drains connections.

> A question describes two VPCs in different accounts that must exchange traffic for one specific API. Both use the 10.0.0.0/16 range. The provider must expose only that API and nothing else.
>
> Options include VPC peering, Transit Gateway, PrivateLink, and a shared VPC via RAM.

What single clause decides this?

A The overlapping CIDR ranges, which eliminate every routing-based option

B The different accounts, since peering does not work across accounts

C The number of VPCs, since two VPCs always means peering

D The requirement to expose one API, since only Transit Gateway can scope traffic

◆ PAUSE AND DECIDE

What single clause decides this?

A Overlapping CIDRs

Peering, Transit Gateway and shared VPCs all build routes, and routes cannot exist between two networks using the same addresses. PrivateLink does not route.

B Different accounts

Peering works across accounts perfectly well. That clause eliminates nothing.

C Two VPCs means peering

A reasonable default that the overlapping ranges override completely. Never apply the default before checking the constraints.

D Exposing one API

A genuine reason to prefer PrivateLink, and it is the second reason here. The overlap eliminates the alternatives outright.

THE SIGNAL

One clause can eliminate three options at once. Scanning for the disqualifying constraint is faster than evaluating each option in turn.

> A question describes needing to know both which resources are shared outside the organisation, and which IAM roles have gone unused for 90 days. Both answers must cover all accounts.
>
> Options offer one organisation-level analyzer, two separate analyzers, credential reports, and Config rules.

What is being tested?

A That IAM Access Analyzer is three separate analyzer types, each created explicitly

B That credential reports cover roles as well as users

C That Config rules can compute unused access

D That external access findings require Security Hub

◆ PAUSE AND DECIDE

What is being tested?

A Three separate analyzers

External access, unused access and internal access are distinct analyzers. Creating one produces none of the others' findings.

B Credential reports cover roles

They cover users and their credentials, per account. Roles and organisation-wide coverage are both outside their scope.

C Config computes unused access

Config evaluates resource configuration against rules. It does not analyse access patterns or compute what went unused.

D Security Hub required

Invented dependency. Access Analyzer produces findings independently, and external access analysis carries no additional charge.

THE SIGNAL

One service name covering several distinct products is its own confusion category, alongside Security Hub CSPM and the three CloudWatch network monitors.

> A question describes an application that must be updated with no capacity reduction. It must pause for review partway through, and reverse automatically if error rates rise. It runs on an Auto Scaling group defined in CloudFormation.
>
> Options mention rolling updates, instance refresh, blue/green, and canary.

What should you recognise here?

A That instance refresh becomes rolling, immutable or canary depending purely on configuration

B That blue/green is always the safest deployment strategy

C That rolling updates cannot maintain capacity under any configuration

D That canary deployments require a load balancer

◆ PAUSE AND DECIDE

What should you recognise here?

A One feature, three strategies by configuration

MinHealthyPercentage 100 keeps capacity, checkpoints give the review pause, and an alarm specification gives automatic reversal. Three requirements, one feature.

B Blue/green is always safest

Over-engineering, which this course has repeatedly shown is a wrong answer rather than a safe one. It also costs a second full environment.

C Rolling can never maintain capacity

Too absolute. Rolling with an additional batch maintains capacity, and instance refresh at 100 percent launches before terminating.

D Canary needs a load balancer

True for traffic-weighted canaries. Instance refresh checkpoints give a canary at the instance level with no traffic splitting.

THE SIGNAL

Not every question is a choice between services. Sometimes one feature answers all the requirements, and the skill is recognising that.

> A question describes a fleet where a configuration file must be present and correct on every instance, including instances launched next month. It must be corrected automatically if someone edits it.
>
> Options include Run Command, State Manager, an Automation runbook, and a Config rule with remediation.

Which discriminator applies?

A. Once now versus continuously enforced, which is Run Command versus State Manager

B. Inside the OS versus on the AWS resource, which is Run Command versus Automation

C. Preventive versus detective, which is Hooks versus Config

D. Known cause versus unknown cause, which is runbook versus DevOps Agent

◆ PAUSE AND DECIDE

Which discriminator applies?

A Once now versus continuously enforced

Two clauses demand continuous enforcement: instances launched later must be covered, and drift must be corrected. That is State Manager, not Run Command.

B Inside the OS versus AWS resource

A real discriminator, and it separates Run Command from Automation. Both act once, so it does not address the continuous requirement.

C Preventive versus detective

Applies to Hooks and Config rules. A Config rule with remediation could work, and it evaluates state rather than enforcing a file's contents.

D Known versus unknown cause

Separates a runbook from an investigation. Nothing here is being investigated; the desired state is already known.

THE SIGNAL

Two clauses point the same way: future instances, and automatic correction. Either alone would suggest continuous enforcement; together they settle it.

> A question describes a Lambda function behind API Gateway that returns throttling errors during a traffic spike. CloudWatch shows concurrency peaked at 280 against an account limit of 1,000, with no other function busy.
>
> Options include raising the account quota, reserved concurrency, provisioned concurrency, and accepting the scaling rate.

Which discriminator applies?

A. Pool limit versus scaling rate: were you throttled with capacity still free?

B. Reserved versus provisioned concurrency: throttling or slow first requests?

C. Synchronous versus asynchronous invocation

D. Destinations versus a dead-letter queue

◆ PAUSE AND DECIDE

Which discriminator applies?

A Pool limit versus scaling rate

280 of 1,000 with nothing else running means the pool was 72 percent free. Only the scaling rate explains throttling with capacity available.

B Reserved versus provisioned

The right discriminator for a different question. Neither raises the rate at which new execution environments are created.

C Sync versus async

Decides retry behaviour on failure. The failure here is throttling at admission, not an error after invocation.

D Destinations versus DLQ

Both handle failed asynchronous events. This function is invoked synchronously through API Gateway.

THE SIGNAL

The numbers are the whole question. Two limits exist, only one appears on the dashboard, and the gap between them is where this failure lives.

> A question describes users behind a load balancer who connect successfully and then see the connection hang. A custom network ACL was recently applied to the subnet. Security groups permit all outbound traffic.
>
> Options include the security group, the NACL's ephemeral range, the idle timeout, and the deregistration delay.

Which discriminator applies?

A. Stateful versus stateless: is return traffic the problem?

B. 502 versus 504: did a target answer badly, or not in time?

C. Cross-zone on or off

D. Evaluate target health versus a separate health check

◆ PAUSE AND DECIDE

Which discriminator applies?

A Stateful versus stateless

Connects then hangs is the stateless signature. A NACL evaluates the response separately, and it leaves on the client's ephemeral port.

B 502 versus 504

Useful when a status code is quoted. None is here, and the symptom described is a hang rather than a returned error.

C Cross-zone

Produces uneven load across zones, not a hang. It also does not interact with NACLs.

D Evaluate target health

Concerns Route 53 alias records. Nothing in the scenario involves DNS.

THE SIGNAL

"A custom NACL was recently applied" plus "connects then hangs" is the pairing. Either detail alone is suggestive; together they are conclusive.

The five things to carry forward

- **Find the one question.** Every pair reduces to a single discriminating property, and the question gets you there faster than the comparison.
- **Trios are harder than pairs.** Health check means replace it, stop sending to it, or stop resolving to it.
- **One word, several meanings.** Endpoint, policy, grant, warm and rotation all mean different things in different places.
- **Scan for the disqualifying clause first.** One constraint often eliminates three options at once.
- **Sometimes the answer is both, and sometimes it is one feature.** Not every question is a choice between services.

> ON THE EXAM
>
> Next, and last: exam mechanics - reading the question, the qualifier words, and managing 130 minutes.